Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Lemos Jerseys

Physical address of dairy:

5712 N Hickman RD	Denair	Stanislaus	95316
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 06/01/1990

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X019-X041-X020-XXXX

B. OPERATORS

LEMOS, NEVIN

Operator name: LEMOS, NEVIN Telephone no.: Landline: (209) 485-5105 Cellular

P.O. Box 235	WATERFORD	CA	95386
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

Prock, Ray S Sr

Operator name: Prock, Ray S Sr Telephone no.: Landline: (209) 667-4693 Cellular: (209) 652-5348

5712 N Hickman RD	Denair	CA	95316
Mailing Address Number and Street	City	State	Zip Code

C. OWNERS

Prock, Ray S Sr

Legal owner name: Prock, Ray S Sr Telephone no.: Landline: (209) 667-4693 Cellular: (209) 652-5348

5712 N Hickman RD	Denair	CA	95316
Mailing Address Number and Street	City	State	Zip Code

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	125	50	0
Number under roof	520	60	50	0	0	0
Maximum number	520	60	50	125	50	0
Average number	520	60	50	125	50	0
Avg live weight (lbs)	1,100	1,150	1,000	800		

Predominant milk cow breed: Jersey

Average milk production: 60 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 15,449.76 *tons per reporting period*

Total nitrogen from manure: 196,530.96 *lbs per reporting period*

After ammonia losses (30% loss applied): 137,571.67 *lbs per reporting period*

Total phosphorus from manure: 32,493.89 *lbs per reporting period*

Total potassium from manure: 96,168.40 *lbs per reporting period*

Total salt from manure: 258,639.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 8,280,000 *gallons*

Total nitrogen generated: 25,530.44 *lbs*

Total phosphorus generated: 3,803.03 *lbs*

Total potassium generated: 49,786.48 *lbs*

Total salt generated: 244,193.89 *lbs*

	8,280,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	8,280,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
TID Irr. Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field #1	26	26	2	both	X019-X041-X020-XXXX
Field #2	9	9	2	both	X019-X041-X020-XXXX
Field #3	28	28	2	both	X019-X041-X020-XXXX
Field #4	16	16	2	both	X019-X041-X020-XXXX
Field #5	11	11	2	both	X019-X041-X020-XXXX
Field #6	2	2	2	both	X019-X041-X020-XXXX
Field #7	30	30	2	both	X019-X041-X020-XXXX
Field #P3	13	13	0	none	X019-X041-X020-XXXX
Field #P5	15	15	0	none	X019-X041-X020-XXXX
Field #P6	8	8	0	none	X019-X041-X020-XXXX
Hill Field	19	19	0	none	X019-X041-X020-XXXX
Pump Field	13	13	0	none	X019-X041-X020-XXXX
Totals for areas that were used for application	122	122	14		
Totals for areas that were not used for application	68	68	0		
Land application area totals	190	190	14		

B. CROPS AND HARVESTS

Field #1

Field name: Field #1

10/19/2022: Winter Forage

Crop: Winter Forage Acres planted: 26 Plant date: 10/19/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	439.40 *ton*	Dry-weight		62.8	18,400.00	2,820.00	18,000.00		9.38

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	219.20	38.40	211.20	0.00
Total actual harvest content	16.90	231.35	35.46	226.32	1,179.40

Field #1

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 26 Plant date: 06/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/16/2023	707.20 *ton*	Dry-weight		67.2	4,600.00	700.00	6,600.00		2.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	27.20	82.08	12.49	117.77	392.55

Field #2

Field name: Field #2

10/19/2022: Winter Forage

Crop: Winter Forage Acres planted: 9 Plant date: 10/19/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	152.10 *ton*	Dry-weight		62.8	18,400.00	2,820.00	18,000.00		9.38

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	219.20	38.40	211.20	0.00
Total actual harvest content	16.90	231.35	35.46	226.32	1,179.40

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 9 Plant date: 06/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/16/2023	244.80 *ton*	Dry-weight		67.2	4,600.00	700.00	6,600.00		2.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	27.20	82.08	12.49	117.77	392.55

Field #3

Field name: Field #3

10/19/2022: Winter Forage

Crop: Winter Forage Acres planted: 28 Plant date: 10/19/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	473.20 *ton*	Dry-weight		54.8	14,900.00	23,700.00	17,400.00		9.23

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	216.00	33.60	249.60	0.00
Total actual harvest content	16.90	227.64	362.08	265.83	1,410.12

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 28 Plant date: 06/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/16/2023	761.60 *ton*	Dry-weight		68.5	4,400.00	900.00	7,000.00		2.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	27.20	75.40	15.42	119.95	376.99

Field #4

Field name: Field #4

10/19/2022: Winter Forage

Crop: Winter Forage Acres planted: 16 Plant date: 10/19/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	270.40 *ton*	Dry-weight		54.8	14,900.00	2,370.00	17,400.00		9.23

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	216.00	33.60	249.60	0.00
Total actual harvest content	16.90	227.64	36.21	265.83	1,410.12

Field #4

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 16 Plant date: 06/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/16/2023	435.20 *ton*	Dry-weight		68.3	4,100.00	1,000.00	4,800.00		1.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	27.20	70.70	17.24	82.78	327.65

Field #5

Field name: Field #5

10/19/2022: Winter Forage

Crop: Winter Forage Acres planted: 11 Plant date: 10/19/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/23/2023	185.90 *ton*	Dry-weight		71.6	27,000.00	5,500.00	79,000.00		12.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	229.50	46.50	540.00	0.00
Total actual harvest content	16.90	259.18	52.80	758.34	1,219.10

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 11 Plant date: 06/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/15/2023	299.20 *ton*	Dry-weight		68.3	4,100.00	1,000.00	4,800.00		1.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	27.20	70.70	17.24	82.78	327.65

Field #6

Field name: Field #6

10/19/2022: Winter Forage

Crop: Winter Forage Acres planted: 2 Plant date: 10/19/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	33.80 *ton*	Dry-weight		62.8	18,400.00	2,820.00	18,000.00		9.38

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	219.20	38.40	211.20	0.00
Total actual harvest content	16.90	231.35	35.46	226.32	1,179.40

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 2 Plant date: 06/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/15/2023	54.40 *ton*	Dry-weight		68.3	4,100.00	1,000.00	4,800.00		1.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	27.20	70.70	17.24	82.78	327.65

Field #7

Field name: Field #7

10/19/2022: Winter Forage

Crop: Winter Forage Acres planted: 30 Plant date: 10/19/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/23/2023	507.00 *ton*	Dry-weight		71.6	27,000.00	5,500.00	79,000.00		12.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	229.50	46.50	540.00	0.00
Total actual harvest content	16.90	259.18	52.80	758.34	1,219.10

Field #7

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 06/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/15/2023	816.00 *ton*	Dry-weight		67.7	4,400.00	1,100.00	4,400.00		1.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	27.20	77.31	19.33	77.31	263.57

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field #1 - 10/19/2022: Winter Forage

Field name: Field #1

Crop: Winter Forage — Plant date: 10/19/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MANURE APPLICATION	Corral solids	164.67	177.86	248.66	3,735.89	130.00 *ton*
Application event totals		164.67	177.86	248.66	3,735.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	62.37	11.18	143.23	831.55	510,000.00 *gal*
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	1,530,000.00 *gal*
Application event totals		62.37	11.18	143.23	831.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	54.70	7.79	106.15	418.08	390,000.00 *gal*
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	1,530,000.00 *gal*
Application event totals		54.70	7.79	106.15	418.08	

Field #1 - 06/12/2023: Corn, silage

Field name: Field #1

Crop: Corn, silage — Plant date: 06/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field #1 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	3,360,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	50.49	7.19	97.98	385.92	360,000.00 *gal*
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	2,970,000.00 *gal*
Application event totals		50.49	7.19	97.98	385.92	

07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	2,700,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	20.26	2.46	30.04	177.17	240,000.00 *gal*
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	2,700,000.00 *gal*
Application event totals		20.26	2.46	30.04	177.17	

08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	2,700,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Field #1 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	20.26	2.46	30.04	177.17	240,000.00 *gal*
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	2,700,000.00 *gal*
Application event totals		20.26	2.46	30.04	177.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	2,700,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Field #2 - 10/19/2022: Winter Forage

Field name: Field #2

Crop: Winter Forage

Plant date: 10/19/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MANURE APPLICATION	Corral solids	164.67	177.86	248.66	3,735.89	45.00 *ton*
Application event totals		164.67	177.86	248.66	3,735.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	63.59	11.40	146.04	847.85	180,000.00 *gal*
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	540,000.00 *gal*
Application event totals		63.59	11.40	146.04	847.85	

Field #2 - 10/19/2022: Winter Forage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	54.70	7.79	106.15	418.08	135,000.00 *gal*
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	540,000.00 *gal*
Application event totals		54.70	7.79	106.15	418.08	

Field #2 - 06/12/2023: Corn, silage

Field name: Field #2

Crop: Corn, silage

Plant date: 06/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	1,488,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	60.78	8.65	117.94	464.54	150,000.00 *gal*
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	1,260,000.00 *gal*
Application event totals		60.78	8.65	117.94	464.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	990,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Field #2 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	32.92	4.01	48.82	287.90	135,000.00 *gal*
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	990,000.00 *gal*
Application event totals		32.92	4.01	48.82	287.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	990,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	990,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	990,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Field #3 - 10/19/2022: Winter Forage

Field name: Field #3

Crop: Winter Forage

Plant date: 10/19/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MANURE APPLICATION	Corral solids	164.67	177.86	248.66	3,735.89	140.00 *ton*
Application event totals		164.67	177.86	248.66	3,735.89	

Field #3 - 10/19/2022: Winter Forage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	57.91	10.38	133.00	772.15	510,000.00 *gal*
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	1,530,000.00 *gal*
Application event totals		57.91	10.38	133.00	772.15	

04/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	54.70	7.79	106.15	418.08	420,000.00 *gal*
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	1,530,000.00 *gal*
Application event totals		54.70	7.79	106.15	418.08	

Field #3 - 06/12/2023: Corn, silage

Field name: Field #3

Crop: Corn, silage

Plant date: 06/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	3,600,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	46.89	6.67	90.98	358.36	360,000.00 *gal*
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	3,240,000.00 *gal*
Application event totals		46.89	6.67	90.98	358.36	

Field #3 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	2,925,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	32.92	4.01	48.82	287.90	420,000.00 *gal*
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	2,925,000.00 *gal*
Application event totals		32.92	4.01	48.82	287.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	2,925,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	18.81	2.29	27.90	164.52	240,000.00 *gal*
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	2,925,000.00 *gal*
Application event totals		18.81	2.29	27.90	164.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	2,925,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Field #4 - 10/19/2022: Winter Forage

Field name: Field #4

Crop: Winter Forage

Plant date: 10/19/2022

Field #4 - 10/19/2022: Winter Forage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MANURE APPLICATION	Corral solids	164.67	177.86	248.66	3,735.89	80.00 *ton*
Application event totals		164.67	177.86	248.66	3,735.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	62.60	11.22	143.76	834.60	315,000.00 *gal*
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	945,000.00 *gal*
Application event totals		62.60	11.22	143.76	834.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	54.70	7.79	106.15	418.08	240,000.00 *gal*
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	945,000.00 *gal*
Application event totals		54.70	7.79	106.15	418.08	

Field #4 - 06/12/2023: Corn, silage

Field name: Field #4

Crop: Corn, silage

Plant date: 06/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	2,112,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Field #4 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	54.70	7.79	106.15	418.08	240,000.00 *gal*
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	1,890,000.00 *gal*
Application event totals		54.70	7.79	106.15	418.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	1,710,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	32.92	4.01	48.82	287.90	240,000.00 *gal*
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	1,710,000.00 *gal*
Application event totals		32.92	4.01	48.82	287.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	1,710,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	1,710,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	1,710,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Field #5 - 10/19/2022: Winter Forage

Field name: Field #5

Crop: Winter Forage

Plant date: 10/19/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MANURE APPLICATION	Corral solids	164.67	177.86	248.66	3,735.89	55.00 *ton*
Application event totals		164.67	177.86	248.66	3,735.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	65.03	11.66	149.36	867.12	225,000.00 *gal*
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	675,000.00 *gal*
Application event totals		65.03	11.66	149.36	867.12	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	54.70	7.79	106.15	418.08	165,000.00 *gal*
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	675,000.00 *gal*
Application event totals		54.70	7.79	106.15	418.08	

Field #5 - 06/12/2023: Corn, silage

Field name: Field #5

Crop: Corn, silage

Plant date: 06/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	1,728,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Field #5 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	1,530,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	47.89	5.83	71.01	418.77	240,000.00 *gal*
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	1,215,000.00 *gal*
Application event totals		47.89	5.83	71.01	418.77	

07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	1,215,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	32.92	4.01	48.82	287.90	165,000.00 *gal*
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	1,215,000.00 *gal*
Application event totals		32.92	4.01	48.82	287.90	

08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	1,215,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	1,215,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Field #6 - 10/19/2022: Winter Forage

Field name: Field #6

Crop: Winter Forage

Plant date: 10/19/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MANURE APPLICATION	Corral solids	164.67	177.86	248.66	3,735.89	10.00 *ton*
Application event totals		164.67	177.86	248.66	3,735.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	71.54	12.82	164.29	953.83	45,000.00 *gal*
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	135,000.00 *gal*
Application event totals		71.54	12.82	164.29	953.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	54.70	7.79	106.15	418.08	30,000.00 *gal*
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	135,000.00 *gal*
Application event totals		54.70	7.79	106.15	418.08	

Field #6 - 06/12/2023: Corn, silage

Field name: Field #6

Crop: Corn, silage

Plant date: 06/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	480,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Field #6 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	360,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	49.38	6.01	73.23	431.85	45,000.00 *gal*
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	225,000.00 *gal*
Application event totals		49.38	6.01	73.23	431.85	

07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	225,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	32.92	4.01	48.82	287.90	30,000.00 *gal*
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	225,000.00 *gal*
Application event totals		32.92	4.01	48.82	287.90	

08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	255,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	255,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Field #7 - 10/19/2022: Winter Forage

Field name: Field #7

Crop: Winter Forage

Plant date: 10/19/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MANURE APPLICATION	Corral solids	164.67	177.86	248.66	3,735.89	150.00 *ton*
Application event totals		164.67	177.86	248.66	3,735.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	63.59	11.40	146.04	847.85	600,000.00 *gal*
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	1,800,000.00 *gal*
Application event totals		63.59	11.40	146.04	847.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	54.70	7.79	106.15	418.08	450,000.00 *gal*
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	1,800,000.00 *gal*
Application event totals		54.70	7.79	106.15	418.08	

Field #7 - 06/12/2023: Corn, silage

Field name: Field #7

Crop: Corn, silage

Plant date: 06/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	3,792,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Field #7 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	62.00	8.82	120.30	473.83	510,000.00 *gal*
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	3,420,000.00 *gal*
Application event totals		62.00	8.82	120.30	473.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	3,060,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	3,060,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	32.92	4.01	48.82	287.90	450,000.00 *gal*
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	3,060,000.00 *gal*
Application event totals		32.92	4.01	48.82	287.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	3,060,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

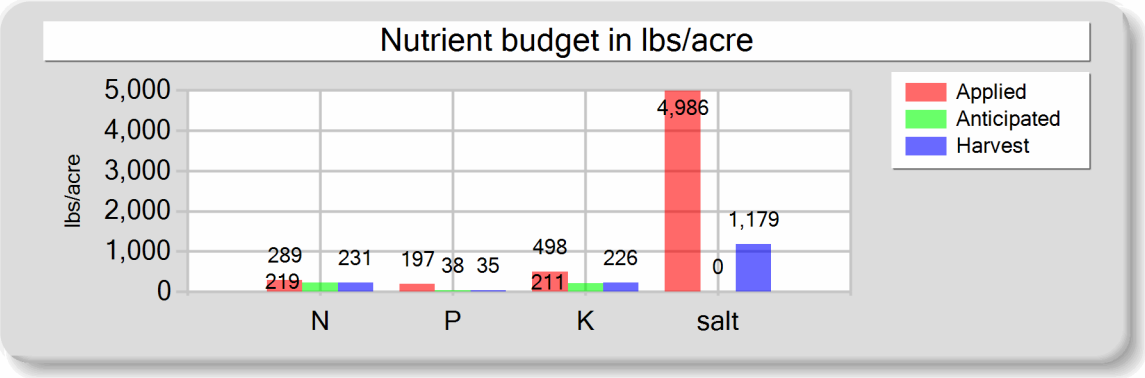
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Irr. Canal	Surface water	0.00	0.00	0.00	0.00	3,060,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

B. NUTRIENT BUDGET

Field #1 - 10/19/2022: Winter Forage

Field name: Field #1 Crop: Winter Forage Plant date: 10/19/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	164.67	177.86	248.66	3,735.89
Process wastewater	117.07	18.97	249.38	1,249.63
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	288.74	196.82	498.04	4,985.52
Anticipated crop nutrient removal	219.20	38.40	211.20	0.00
Actual crop nutrient removal	231.35	35.46	226.32	1,179.40
Nutrient balance	57.38	161.37	271.71	3,806.12
Applied to removed ratio	1.25	5.55	2.20	4.23

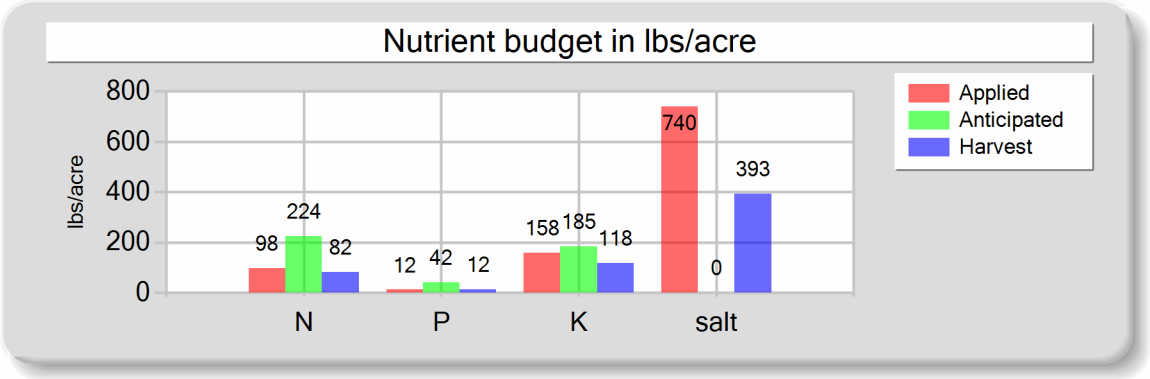
Fresh water applied	
3,060,000.00	*gallons*
112.69	*acre-inches*
4.33	*inches/acre*

Process wastewater applied	
900,000.00	*gallons*
33.14	*acre-inches*
1.27	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #1 - 06/12/2023: Corn, silage

Field name: Field #1 Crop: Corn, silage Plant date: 06/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	91.01	12.12	158.07	740.27
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	98.01	12.12	158.07	740.27
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	82.08	12.49	117.77	392.55
Nutrient balance	15.93	-0.37	40.30	347.72
Applied to removed ratio	1.19	0.97	1.34	1.89

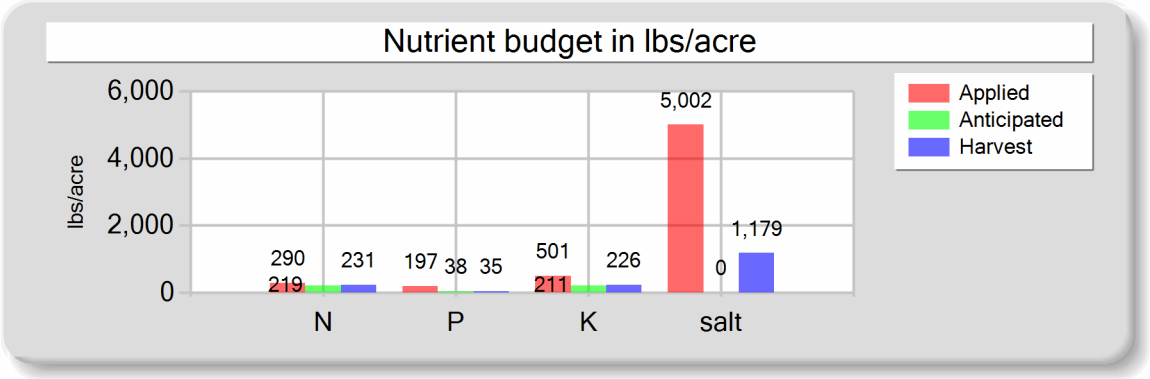
Fresh water applied	
19,830,000.00	*gallons*
730.27	*acre-inches*
28.09	*inches/acre*

Process wastewater applied	
840,000.00	*gallons*
30.93	*acre-inches*
1.19	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #2 - 10/19/2022: Winter Forage

Field name: Field #2 Crop: Winter Forage Plant date: 10/19/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	164.67	177.86	248.66	3,735.89
Process wastewater	118.29	19.19	252.19	1,265.94
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	289.96	197.04	500.85	5,001.83
Anticipated crop nutrient removal	219.20	38.40	211.20	0.00
Actual crop nutrient removal	231.35	35.46	226.32	1,179.40
Nutrient balance	58.60	161.59	274.52	3,822.42
Applied to removed ratio	1.25	5.56	2.21	4.24

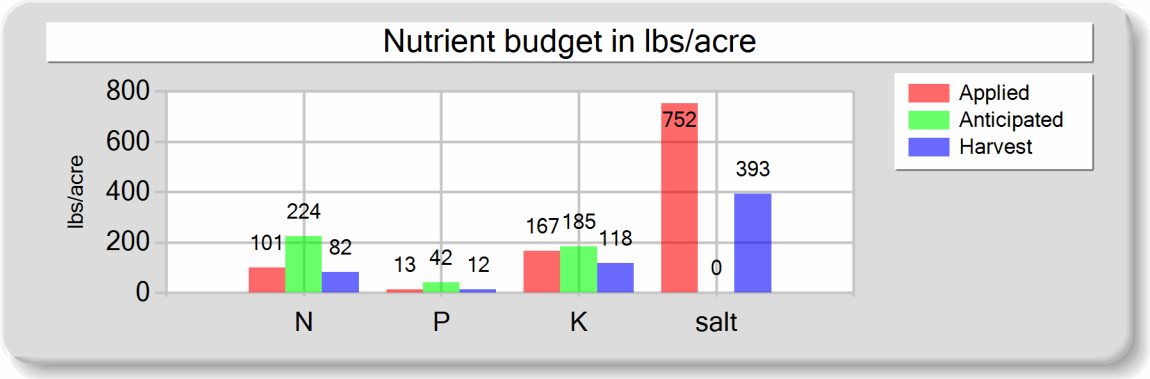
Fresh water applied	
1,080,000.00	*gallons*
39.77	*acre-inches*
4.42	*inches/acre*

Process wastewater applied	
315,000.00	*gallons*
11.60	*acre-inches*
1.29	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #2 - 06/12/2023: Corn, silage

Field name: Field #2 Crop: Corn, silage Plant date: 06/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	93.70	12.66	166.76	752.44
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	100.70	12.66	166.76	752.44
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	82.08	12.49	117.77	392.55
Nutrient balance	18.62	0.17	49.00	359.89
Applied to removed ratio	1.23	1.01	1.42	1.92

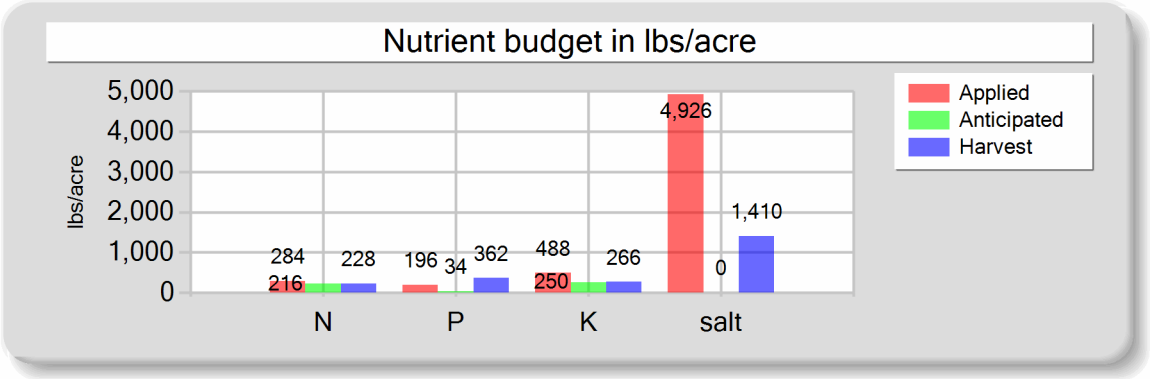
Fresh water applied	
7,698,000.00	*gallons*
283.49	*acre-inches*
31.50	*inches/acre*

Process wastewater applied	
285,000.00	*gallons*
10.50	*acre-inches*
1.17	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #3 - 10/19/2022: Winter Forage

Field name: Field #3 Crop: Winter Forage Plant date: 10/19/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	164.67	177.86	248.66	3,735.89
Process wastewater	112.61	18.17	239.15	1,190.24
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	284.28	196.03	487.81	4,926.13
Anticipated crop nutrient removal	216.00	33.60	249.60	0.00
Actual crop nutrient removal	227.64	362.08	265.83	1,410.12
Nutrient balance	56.65	-166.05	221.98	3,516.00
Applied to removed ratio	1.25	0.54	1.84	3.49

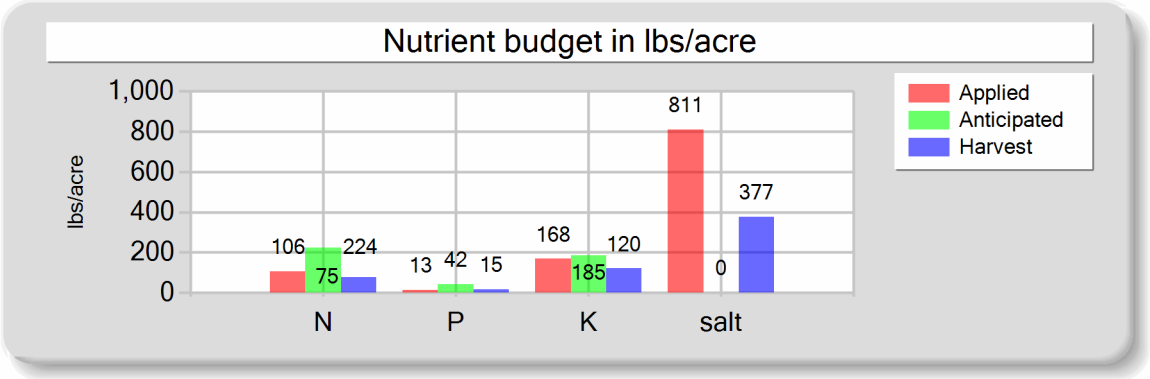
Fresh water applied	
3,060,000.00	*gallons*
112.69	*acre-inches*
4.02	*inches/acre*

Process wastewater applied	
930,000.00	*gallons*
34.25	*acre-inches*
1.22	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #3 - 06/12/2023: Corn, silage

Field name: Field #3 Crop: Corn, silage Plant date: 06/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	98.62	12.97	167.70	810.78
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	105.62	12.97	167.70	810.78
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	75.40	15.42	119.95	376.99
Nutrient balance	30.22	-2.45	47.75	433.78
Applied to removed ratio	1.40	0.84	1.40	2.15

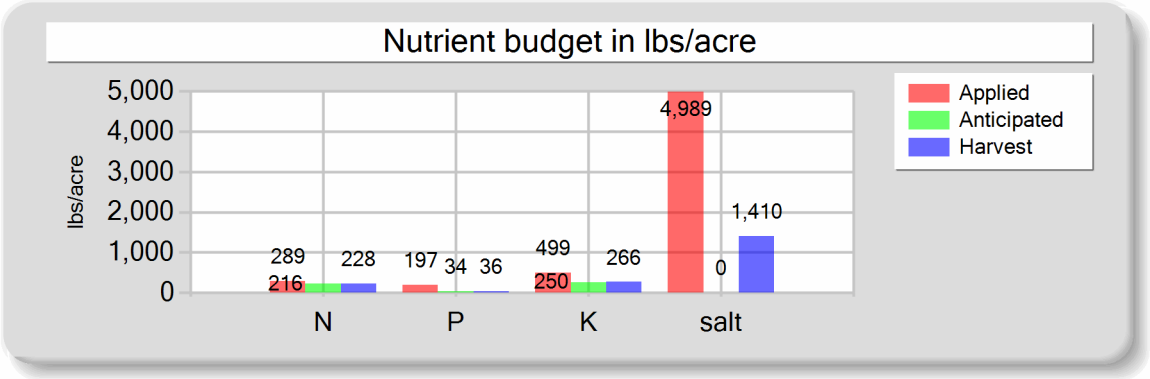
Fresh water applied	
21,465,000.00	*gallons*
790.48	*acre-inches*
28.23	*inches/acre*

Process wastewater applied	
1,020,000.00	*gallons*
37.56	*acre-inches*
1.34	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #4 - 10/19/2022: Winter Forage

Field name: Field #4 Crop: Winter Forage Plant date: 10/19/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	164.67	177.86	248.66	3,735.89
Process wastewater	117.30	19.01	249.90	1,252.69
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	288.97	196.87	498.57	4,988.58
Anticipated crop nutrient removal	216.00	33.60	249.60	0.00
Actual crop nutrient removal	227.64	36.21	265.83	1,410.12
Nutrient balance	61.33	160.66	232.74	3,578.46
Applied to removed ratio	1.27	5.44	1.88	3.54

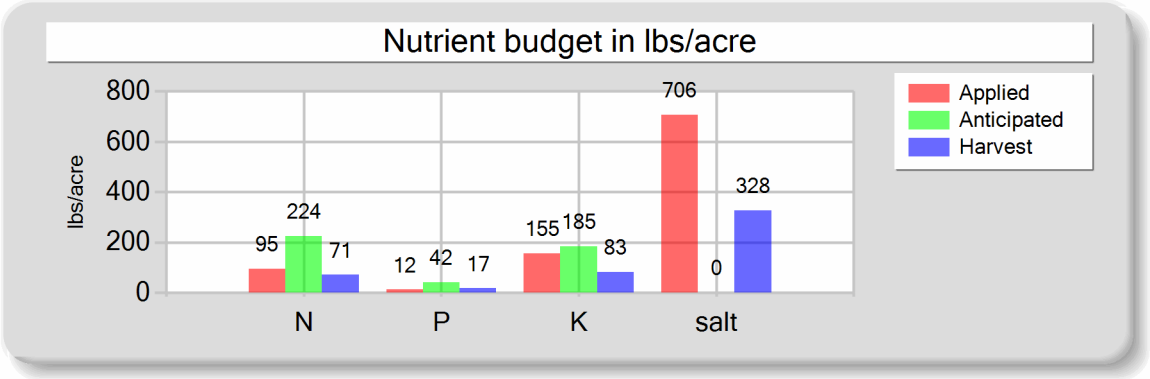
Fresh water applied	
1,890,000.00	*gallons*
69.60	*acre-inches*
4.35	*inches/acre*

Process wastewater applied	
555,000.00	*gallons*
20.44	*acre-inches*
1.28	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #4 - 06/12/2023: Corn, silage

Field name: Field #4 Crop: Corn, silage Plant date: 06/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	87.62	11.79	154.97	705.99
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	94.62	11.79	154.97	705.99
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	70.70	17.24	82.78	327.65
Nutrient balance	23.92	-5.45	72.19	378.34
Applied to removed ratio	1.34	0.68	1.87	2.15

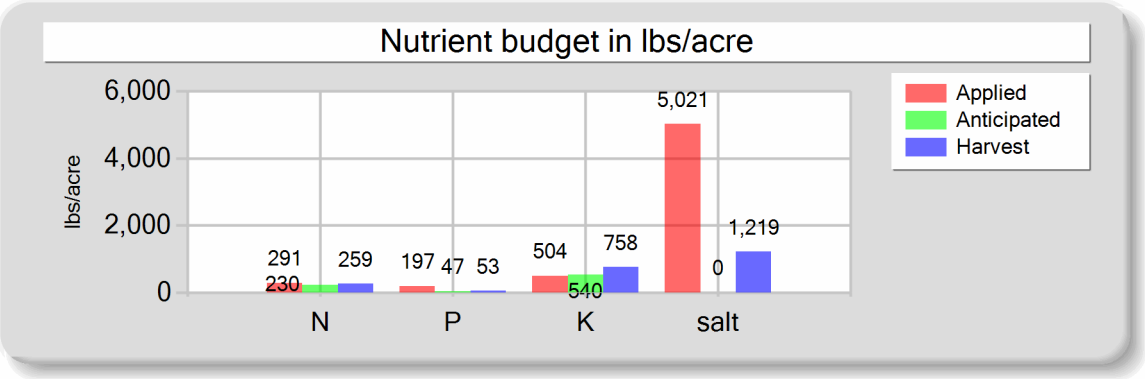
Fresh water applied	
12,552,000.00	*gallons*
462.25	*acre-inches*
28.89	*inches/acre*

Process wastewater applied	
480,000.00	*gallons*
17.68	*acre-inches*
1.10	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #5 - 10/19/2022: Winter Forage

Field name: Field #5 Crop: Winter Forage Plant date: 10/19/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	164.67	177.86	248.66	3,735.89
Process wastewater	119.74	19.44	255.50	1,285.21
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	291.40	197.30	504.17	5,021.10
Anticipated crop nutrient removal	229.50	46.50	540.00	0.00
Actual crop nutrient removal	259.18	52.80	758.34	1,219.10
Nutrient balance	32.23	144.51	-254.17	3,802.00
Applied to removed ratio	1.12	3.74	0.66	4.12

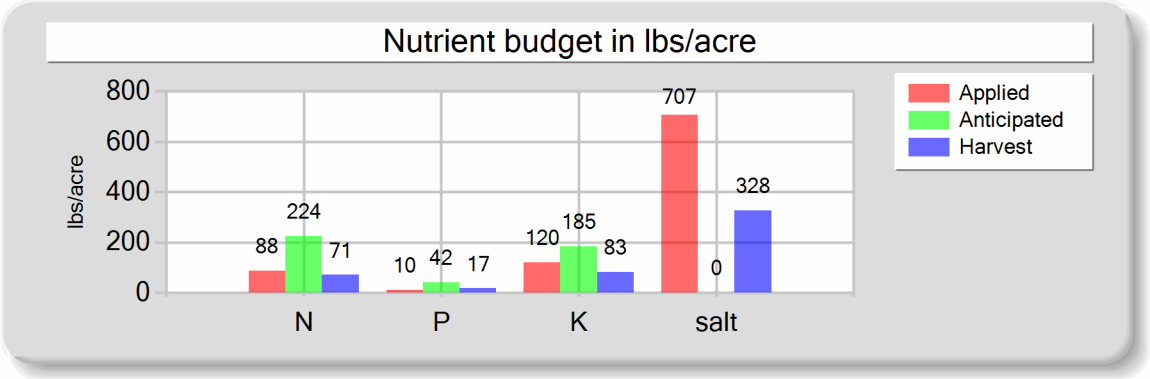
Fresh water applied	
1,350,000.00	*gallons*
49.72	*acre-inches*
4.52	*inches/acre*

Process wastewater applied	
390,000.00	*gallons*
14.36	*acre-inches*
1.31	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #5 - 06/12/2023: Corn, silage

Field name: Field #5 Crop: Corn, silage Plant date: 06/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	80.81	9.83	119.83	706.67
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	87.81	9.83	119.83	706.67
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	70.70	17.24	82.78	327.65
Nutrient balance	17.10	-7.41	37.05	379.02
Applied to removed ratio	1.24	0.57	1.45	2.16

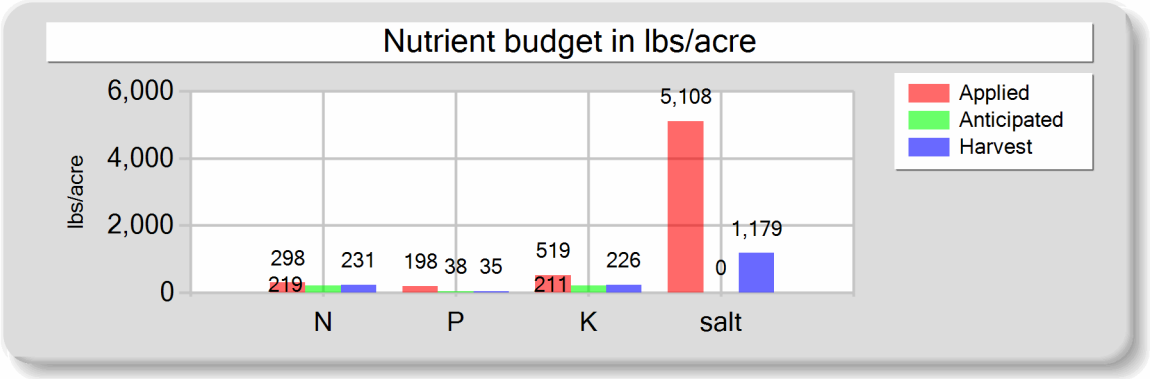
Fresh water applied	
9,333,000.00	*gallons*
343.70	*acre-inches*
31.25	*inches/acre*

Process wastewater applied	
405,000.00	*gallons*
14.91	*acre-inches*
1.36	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #6 - 10/19/2022: Winter Forage

Field name: Field #6 Crop: Winter Forage Plant date: 10/19/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	164.67	177.86	248.66	3,735.89
Process wastewater	126.24	20.61	270.44	1,371.92
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	297.91	198.47	519.10	5,107.81
Anticipated crop nutrient removal	219.20	38.40	211.20	0.00
Actual crop nutrient removal	231.35	35.46	226.32	1,179.40
Nutrient balance	66.55	163.01	292.78	3,928.40
Applied to removed ratio	1.29	5.60	2.29	4.33

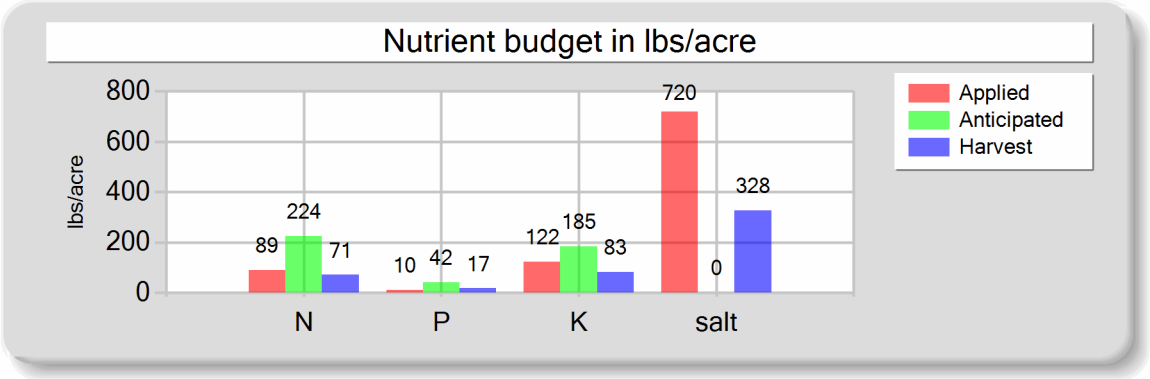
Fresh water applied	
270,000.00	*gallons*
9.94	*acre-inches*
4.97	*inches/acre*

Process wastewater applied	
75,000.00	*gallons*
2.76	*acre-inches*
1.38	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #6 - 06/12/2023: Corn, silage

Field name: Field #6 Crop: Corn, silage Plant date: 06/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	82.30	10.01	122.05	719.76
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	89.30	10.01	122.05	719.76
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	70.70	17.24	82.78	327.65
Nutrient balance	18.60	-7.23	39.27	392.11
Applied to removed ratio	1.26	0.58	1.47	2.20

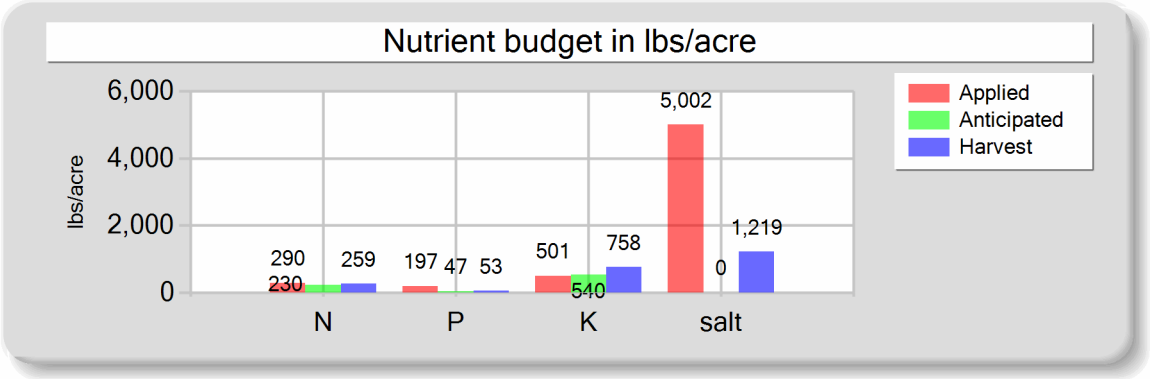
Fresh water applied	
2,025,000.00	*gallons*
74.57	*acre-inches*
37.29	*inches/acre*

Process wastewater applied	
75,000.00	*gallons*
2.76	*acre-inches*
1.38	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #7 - 10/19/2022: Winter Forage

Field name: Field #7 Crop: Winter Forage Plant date: 10/19/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	164.67	177.86	248.66	3,735.89
Process wastewater	118.29	19.19	252.19	1,265.94
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	289.96	197.04	500.85	5,001.83
Anticipated crop nutrient removal	229.50	46.50	540.00	0.00
Actual crop nutrient removal	259.18	52.80	758.34	1,219.10
Nutrient balance	30.78	144.25	-257.49	3,782.73
Applied to removed ratio	1.12	3.73	0.66	4.10

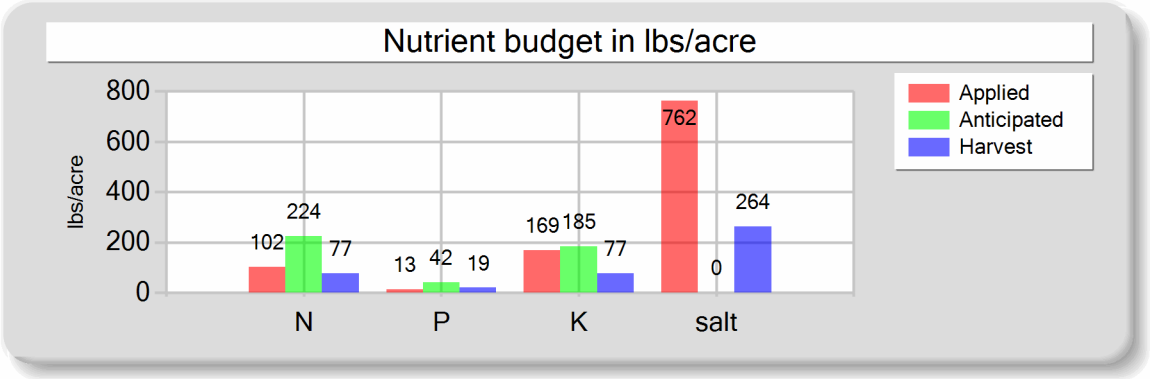
Fresh water applied	
3,600,000.00	*gallons*
132.58	*acre-inches*
4.42	*inches/acre*

Process wastewater applied	
1,050,000.00	*gallons*
38.67	*acre-inches*
1.29	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #7 - 06/12/2023: Corn, silage

Field name: Field #7 Crop: Corn, silage Plant date: 06/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	94.92	12.83	169.12	761.73
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	101.92	12.83	169.12	761.73
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	77.31	19.33	77.31	263.57
Nutrient balance	24.60	-6.50	91.81	498.16
Applied to removed ratio	1.32	0.66	2.19	2.89

Fresh water applied	
22,512,000.00	*gallons*
829.04	*acre-inches*
27.63	*inches/acre*

Process wastewater applied	
960,000.00	*gallons*
35.35	*acre-inches*
1.18	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

57946301

Sample and source description: 57946301
Sample date: 10/07/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 14.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,350.00	20,900.00	29,220.00	23,450.00	9,850.00	4,550.00	0.00	0.00		43.90
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00		1.00

T3132021B

Sample and source description: T3132021B
Sample date: 05/12/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 8.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,000.00	8,300.00	8,700.00							0.00
DL	1.00	1.00	1.00							1.00

T3342028B

Sample and source description: T3342028B
Sample date: 12/08/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 17.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,800.00	10,700.00	17,000.00	32,300.00	11,300.00	5,170.00	4,150.00	2,140.00		37.80
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00		1.00

B. PROCESS WASTEWATER ANALYSES

T3053006A

Sample and source description: T3053006A

Sample date: 02/22/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	381.00	213.00	0.00	0.00	68.30	875.00								5,830.00	5,080
DL	1.00	1.00	1.00	1.00	1.00	1.00								100.00	1

T3132021A

Sample and source description: T3132021A

Sample date: 05/12/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	437.00	249.00	0.00	0.00	62.20	848.00								5,740.00	3,340
DL	1.00	1.00	1.00	1.00	1.00	1.00								100.00	1

T3216014A

Sample and source description: T3216014A

Sample date: 08/04/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	263.00	165.00	0.00	0.00	32.00	390.00	63.00	86.30	114.00	1,720.00	0.00	212.00	178.00	3,940.00	2,300
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	2.00	1.00	1.00	100.00	1

T3342028A

Sample and source description: T3342028A

Sample date: 12/08/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	428.00	274.00	0.00	0.00	13.70	651.00								5,320.00	6,920
DL	1.00	1.00	1.00	1.00	1.00	1.00								100.00	1

C. FRESH WATER ANALYSES

TID Irr. Canal

T3235025A

Sample description: T3235025A

Sample date: 08/23/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	0
DL	0.14	0.50	0.14								100.00	1

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Field #1 - 10/19/2022: Winter Forage

T3130045A

Sample and source description: T3130045A

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,400.00	2,820.00	18,000.00		9.38
DL	1.00	1.00	1.00		1.00

Field #1 - 06/12/2023: Corn, silage

Field #1 - 06/12/2023: Corn, silage

61313301

Sample and source description: 61313301

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,600.00	700.00	6,600.00		2.20
DL	1.00	1.00	1.00		1.00

Field #2 - 10/19/2022: Winter Forage

T3130045A

Sample and source description: T3130045A

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,400.00	2,820.00	18,000.00		9.38
DL	1.00	1.00	1.00		1.00

Field #2 - 06/12/2023: Corn, silage

61313301

Sample and source description: 61313301

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,600.00	700.00	6,600.00		2.20
DL	1.00	1.00	1.00		1.00

Field #3 - 10/19/2022: Winter Forage

T3130045B

Sample and source description: T3130045B

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 54.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,900.00	23,700.00	17,400.00		9.23
DL	1.00	1.00	1.00		1.00

Field #3 - 06/12/2023: Corn, silage

61313302

Sample and source description: 61313302

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,400.00	900.00	7,000.00		2.20
DL	1.00	1.00	1.00		1.00

Field #4 - 10/19/2022: Winter Forage

T3130045B

Sample and source description: T3130045B

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 54.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,900.00	2,370.00	17,400.00		9.23
DL	1.00	1.00	1.00		1.00

Field #4 - 06/12/2023: Corn, silage

61313303

Sample and source description: 61313303

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,100.00	1,000.00	4,800.00		1.90
DL	1.00	1.00	1.00		1.00

Field #5 - 10/19/2022: Winter Forage

T3114017A

Sample and source description: T3114017A

Sample date: 04/24/2023 Source of analysis: Other/estimated Method of reporting: Dry-weight

Moisture: 71.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,000.00	5,500.00	79,000.00		12.70
DL	1.00	1.00	1.00		1.00

Field #5 - 06/12/2023: Corn, silage

61313303

Sample and source description: 61313303

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,100.00	1,000.00	4,800.00		1.90
DL	1.00	1.00	1.00		1.00

Field #6 - 10/19/2022: Winter Forage

T3130045A

Sample and source description: T3130045A

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,400.00	2,820.00	18,000.00		9.38
DL	1.00	1.00	1.00		1.00

Field #6 - 06/12/2023: Corn, silage

61313303

Sample and source description: 61313303

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,100.00	1,000.00	4,800.00		1.90
DL	1.00	1.00	1.00		1.00

Field #7 - 10/19/2022: Winter Forage

T3114017A

Sample and source description: T3114017A

Sample date: 04/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,000.00	5,500.00	79,000.00		12.70
DL	1.00	1.00	1.00		1.00

Field #7 - 06/12/2023: Corn, silage

61313304

Sample and source description: 61313304

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,400.00	1,100.00	4,400.00		1.50
DL	1.00	1.00	1.00		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

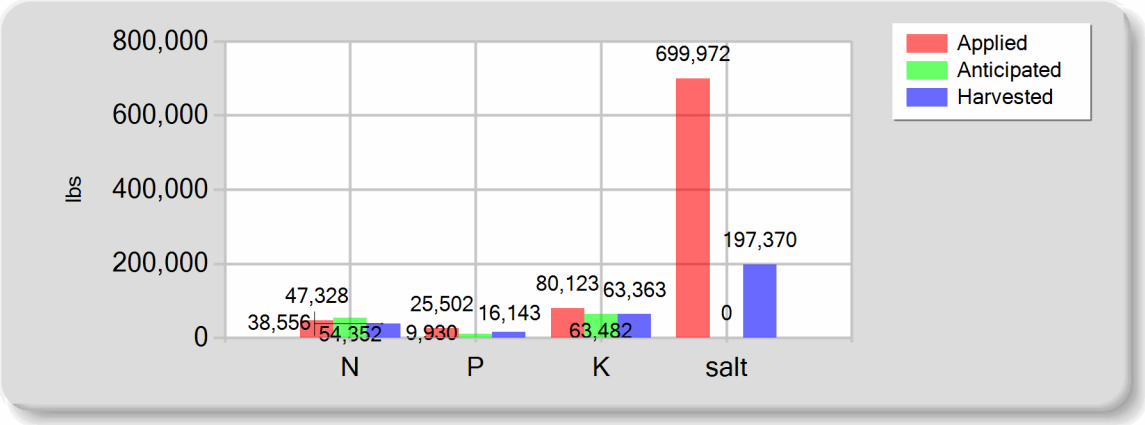
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

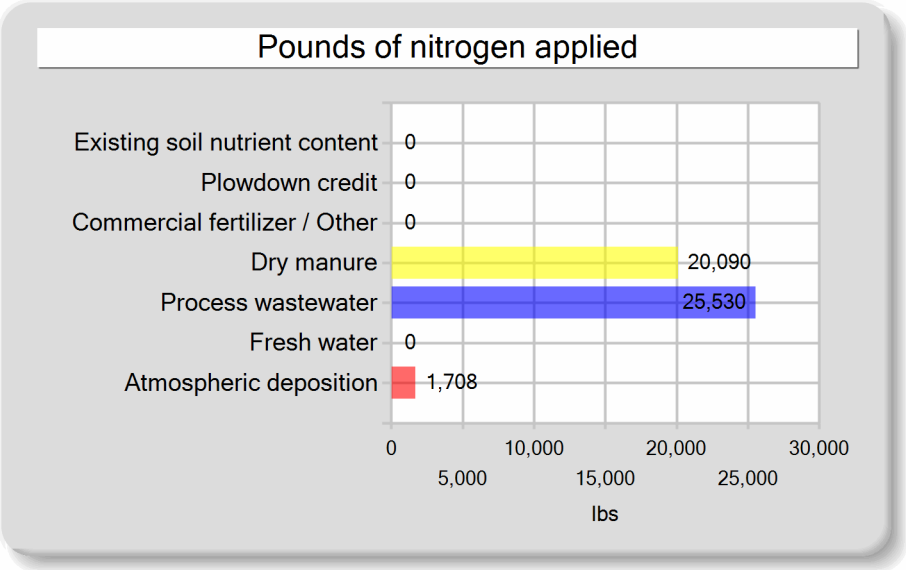
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

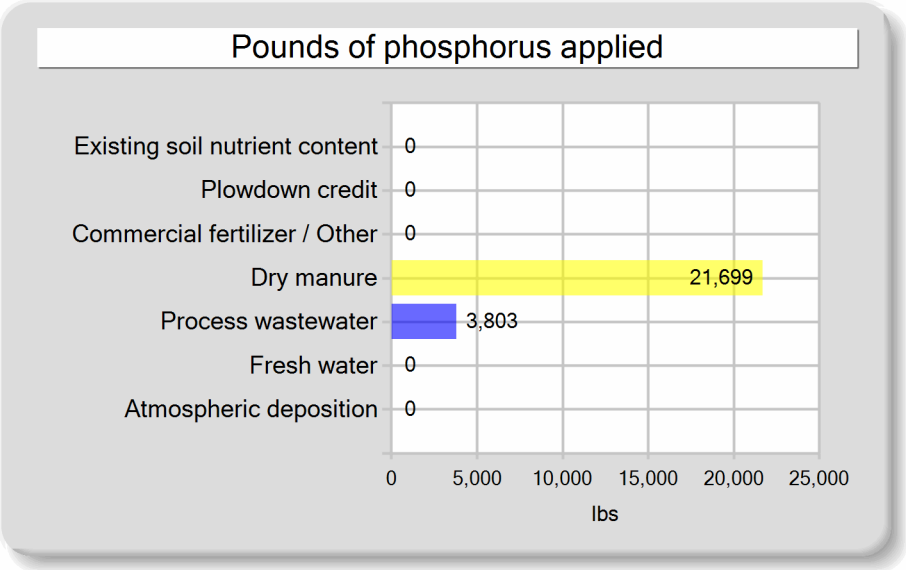
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	20,089.56	21,698.80	30,336.79	455,778.58
Process wastewater	25,530.44	3,803.03	49,786.48	244,193.89
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	1,708.00	0.00	0.00	0.00
Total nutrients applied	47,328.00	25,501.83	80,123.27	699,972.47
Anticipated crop nutrient removal	54,351.90	9,929.70	63,482.40	0.00
Actual crop nutrient removal	38,556.13	16,143.03	63,362.67	197,370.32
Nutrient balance	8,771.87	9,358.80	16,760.60	502,602.15
Applied to removed ratio	1.23	1.58	1.26	3.55

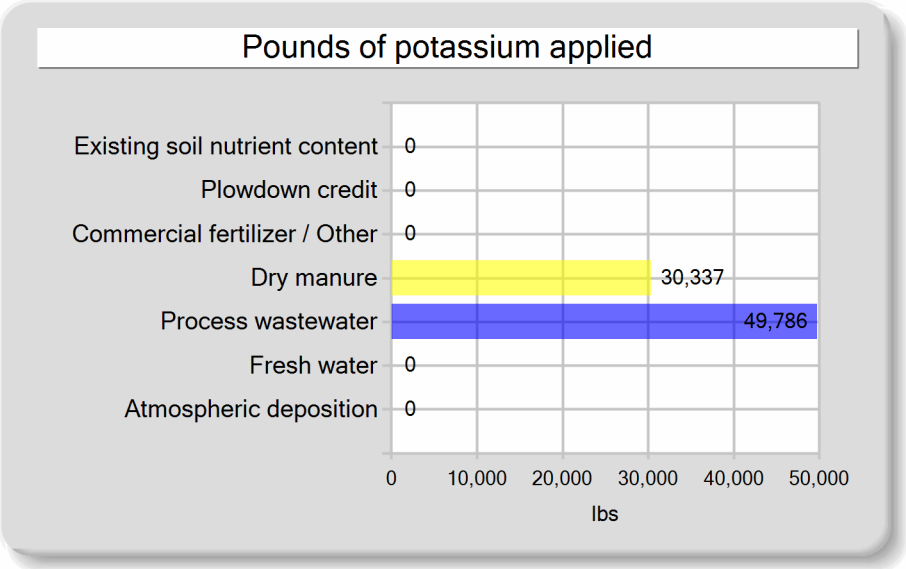
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

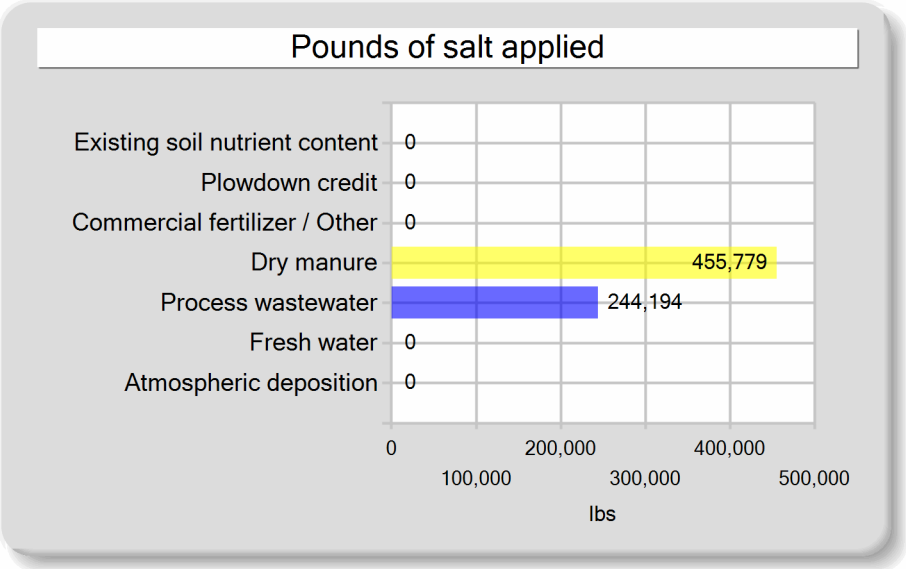


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Ray S. Prock Sr	
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Ray S Prock, Sr	NEVIN LEMOS
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6-19-24	6-19-24
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: LEMOS JERSEYS
Mailing address: P.O. BOX 235
(not facility address) WATERFORD, CA 95386
Phone#: (209) 485-5105

Authorized Copy Released To: JCRK, INC. (209) 327-0992

Sampled By: JACQUIE CAPTEIN
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
H1 EC*, NO3-N, NH4-N*
H6 EC, TN, TDS
H7 Combination of H1 & H6
H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
H3 EC*, NH4-N, TKN, TP, K, TDS
H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
H8 Combination of H3 &H5
H9 Combination of H4 &H5

TILE DRAIN
H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

PLANT TISSUE
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	8/9/23	11:30a	TID Irr Canal	Irr Water	21.0°		X				
2											
3											
4											
5											
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time
1st	Jacquie Capteiu	JCRK, INC.		8/11/23 8:00a
2nd		DENELE ANALYTICAL, INC.	8-23-23 11:00	

(LABORATORY USE ONLY)

Logged In By:	LABORATORY NUMBER
	T3235025

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Lemos Jerseys
P.O. Box 235
Waterford, CA 95386

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 8/23/23 1:00 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 8/25/23

Irrigation/ Ground Water (H1)
SOURCE: Canal Water

NOTES: Temperature was above qualifying at 21.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
TID Irr. Canal	T3235025A	Electrical Conductivity (EC)	EPA 120.1 and	< 0.100	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	16.7	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	101	95.1 (T3235024B)	95.9 (T3235024B)	0.815
Electrical Conductivity (EC)	97.1			4.81
Nitrate Nitrogen (NO3-N)	104			7.24

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